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Food for Thought

Essays on Green Public Procurement and the Market for Organic Food

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Umeå 2021

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Dissertation for PhD

ISBN: 978-91-7855-639-7 (print)

ISBN: 978-91-7855-640-3 (PDF)

ISSN: 0348-1018

Umeå Economic Studies No.996

Electronic version available at: <http://umu.diva-portal.org/>

Cover by Ida Åberg, Inhousebyrå, Umeå University

Cover picture by nld/Shutterstock

Printed by: Print & Media at Umeå University, Sweden 2021

Abstract

Paper [I] In this paper, we study a Green Public Procurement (GPP) policy decided by the Swedish government in 2006, stating objectives related to organic farming. The policy aims to increase the public sector's organic food purchases, in order to incentivise Swedish farmers to convert to organic practices, thereby contributing to national environmental quality objectives. We analyse the effect of organic food procurement on organic agricultural land, using panel data from 2003 to 2016 including information on municipalities' organic food purchases, land use, and direct subsidies aimed at organic production. Based on different specifications and mainly FGLS estimations, we conclude that the 2006 organic food policy is associated with a significant positive impact on organic agricultural land. A significant effect of direct agricultural policy in the form of subsidies is also found.

Paper [II] In this paper, we analyse the uptake of a national and voluntary green public procurement policy among Swedish municipalities. The policy, decided in 2006, was intended to contribute to increased organic farming by stipulating that at least 25 percent of public food purchases be organic. Based on survey data on municipalities' organic food purchases for the period 2003–2016, supplemented with municipal characteristics, we analyse the determinants underpinning uptake of the policy, accounting for potential selection bias. The main finding is that political goals have a significant and positive effect on the share of organic food purchases, suggesting that there is an uptake and that the voluntary policy is in fact implemented. Secondly, we find that the increase in expenditures per capita devoted to organic food is quite substantial following the adoption of an organic food policy.

Paper [III] In this paper, I study consumer demand for organic and conventional milk, using weekly scanner data from the Swedish retail market for the years 2011–2017. Own- and cross-price elasticities of demand are estimated using a quadratic almost ideal demand system. The paper contributes to previous literature by (i) studying a market with relatively small organic price premiums, (ii) using a highly representative sample of retailers, and (iii) differentiating between private labels and brands. Results show that demand for organic milk is relatively elastic, despite relatively small organic price premiums in the Swedish milk market. Results also show that demand for branded products is, generally, less price elastic compared to private label products, suggesting that consumers have strong preferences for traditional, regional brands.

Paper [IV] In this paper, I analyse wholesale to retail price transmission for organic and conventional milk in the Swedish milk sector, using time-series analysis applied to monthly price data for the period Jan 2007–Nov 2017. Although much empirical work addresses the efficiency of food supply chains by studying price transmission, studies on price transmission for quality-differentiated food are scarce, and particularly for organic food vis-à-vis conventional food. This study adds to this scarce literature. Estimations are performed using the non-linear ARDL model which allows for asymmetric cointegration of prices and a simultaneous analysis of short- and long-run asymmetry, the latter of which has been largely overlooked in previous studies. In the case of conventional milk, results indicate positive asymmetries in magnitude both in the short-run and the long-run. For organic milk, the long-run positive asymmetry is less pronounced and organic consumers are therefore likely to experience smaller differences between surplus losses and gains following positive and negative wholesale price changes, respectively.

Keywords: Organic food policy, agricultural policy, Green Public Procurement, environmental policy, consumer demand, demand system, price transmission, survey data, retail scanner data.

Acknowledgements

They say there is only one way to eat an elephant: one bite at a time. Writing a PhD thesis is strikingly similar, with the process consisting of incrementally small steps forward (and often enough backwards). When those forward steps have been v.tiny, I am fortunate to have had the unwavering support of my main supervisor, Prof. Sofia Lundberg. Sofia, thank you for including me in this and other research projects, and for your co-authorship on two of the papers. The clarity and grit you bring into research is highly inspiring, and I look forward to many more collaborations. My assistant supervisor, Prof. Tommy Lundgren, you are a man of few words, so I will honor you by being concise, but nonetheless grateful. Thank you! For his co-authorship on two of the papers, I would also like to thank Pelle Marklund.

The papers in this thesis have greatly benefitted from discussions at internal seminars in the department of Economics. For this I like to extend my deep gratitude to Jurate Jaraite, Golnaz Amjadi, Magnus Wikström, Johan Holmberg, Raoul Theler, David Granlund, Linn Karlsson, Giovanni Forchini, Gauthier Lanot and Sef Meens Eriksson. I am also greatly indebted to my constant discussant, Mattias Vesterberg, for his many careful readings and constructive comments.

If academia is known for its glass ceiling, I am grateful that the department of Economics at least practices a policy of open doors. Thank you to Lars Persson, Niklas Hanes and Gauthier Lanot for always keeping your doors open as heads of department and research. The administrative support from Sara Widmark, and her swift help in matters concerning teaching and course work is gratefully acknowledged. During the past 4 years, I have highly appreciated the company of past and present doctoral students at the department; Golnaz, Alejandro x 2, Aemiro, Victoria, André, Johan H, Johan G, Linn, Sef, Balder, Samielle, Stina and Peter. I could not have asked for a better way to start my PhD studies than to share office with Linn Karlsson. Our conversations on PhD life, teaching, and cat parenting have been thoroughly enjoyed. To the rest of the colleagues at the department of Economics who have contributed to forging a welcoming and inspiring work environment, Thank you!

Funding from the Swedish Competition Authority is gratefully acknowledged. The papers and I have benefitted highly from participating in workshops and conferences around Sweden and Europe. Thanks, in particular, to Joakim Wallenklint at the Swedish Competition Authority for organizing and letting me present at the SWCR, 2017. I am also indebted to HUI research for providing data and administrative support. As a visiting PhD student, I have had great experiences at the Bilbao Center for Climate Change in Bilbao, the Managi Lab at University of Kyushu, and the Economics Unit at Luleå University of Technology. Thank you for offering office space and pleasant company!

I am very grateful for the support from friends and family. A very special s/o to Anna for your set of clear eyes and full heart. Vital reboots of mind and body have taken place on the beachvolley court, and it would not have been as enjoyable without the beach girls: Sara, Nanny-Maja, Stina, Jonna, Carolina, Anniina, Frida - a sincere high-five for your ability to lift spirits. To Jessica, Caroline, Sanna, Steffen, Rebecka, Tove, Christopher, Jennie, Maria, Lisa, thank you for much needed downtime AFK. To my sisters Julia and Linnea, thank you for always offering support and entertainment. Frænka Hanna is looking forward to more time for visits and playdates in the near future! I am very grateful for the perspectives passed on to me by my parents. To my dad, thank you for instilling a healthy dose of critical thinking and, at least, a theoretical interest for farming. To my mum, thank you for your relentless positive mindset and for inspiring to lifelong learning and crossword solving. Last, but not least, I am very grateful for having Mattias in my life. Sharing work and living space with you is a sheer pleasure, and I look forward to all that lies ahead.

Umeå, September 2021

Hanna

This thesis consists of an introduction, a summary of the papers, and the following four self-contained papers:

Paper [I]

Lindström, H., Lundberg, S., Marklund, P-O. (2020). How Green Public Procurement can drive conversion of farmland: An empirical analysis of an organic food policy. *Ecological Economics*, 172:106622. (Reprinted with permission)

Paper [II]

Lindström, H., Lundberg, S., Marklund, P-O. (2021). Green Public Procurement: An empirical analysis of the uptake of organic food policy. *Umeå Economic Studies*, No. 997. Revise and resubmit.

Paper [III]

Lindström, H. (2021). The Swedish consumer market for organic and conventional milk: A demand system analysis. *HFI working paper*, No.21. *Umeå Economic Studies*, No. 998. Revise and resubmit.

Paper [IV]

Lindström, H. (2021). Price transmission for organic and conventional milk products in Sweden. *HFI working paper*, No.22. *Umeå Economic Studies*, No. 999.

1. Introduction

For the past decades, the role of organic food production has been heavily emphasized when it comes to realizing Swedish and European environmental ambitions, and it is central in the recently launched European Green Deal (European Commission, 2004, 2014, 2021; Government communication, 2006; Swedish Government, 2017). The dual potential of organic production is often underscored as it can provide a public good in terms of increased environmental protection and animal welfare, as well as a private good by responding to an increased consumer demand for a more sustainable diet.

Policies within the European Union (EU) aimed at stimulating the organic food production mainly include a coherent EU organic labelling standard, and organic subsidies. The latter are designed and co-financed according to each member state's priorities within the rural development programmes (RDP). Furthermore, the EU also encourages member states to use Green Public Procurement (GPP) policies aimed at public authorities to increase their demand for organic food products, in order to incentivise farmers to convert to organic production (European Commission, 2014). Differentiated VAT rates for organic and conventional food products or input factors have also been discussed as a means to reduce organic price premiums and stimulate demand for organic products (see, e.g., Oosterhuis et al. (2008)). In Sweden, subsidies to organic production have been in force since 1995, and in 2006, the Swedish government also launched a national GPP policy, encouraging the public sector to increase their organic food purchases (Government communication, 2006; Jørgensen, 2001).

However, the effect of GPP as an environmental policy instrument is not self-evident, and depends on, e.g., the public sector's buyer power, private consumers' response, and on the capacity of the public authority to implement the policy (Cheng et al., 2018; Lundberg et al., 2016; Lundberg et al., 2015; Marron, 1997). Among the few empirical studies on the impact of GPP, findings from the Swedish market for cleaning services (Lundberg et al., 2015) and the construction sector in California (Simcoe & Toffel, 2014) are mixed, and studies from the food sector are lacking. Further, private consumers' price sensitivities for organic and conventional food, and the price setting strategies of actors within the supply chain are pivotal for subsidies and taxes to have the intended effect on organic production. Despite a large policy focus on organic production in Sweden, there is scarce empirical knowledge on these factors. The aim of this thesis is to address these gaps in the literature.

This thesis consists of four self-contained papers. Two of the papers relate to the potential of GPP to contribute to increased organic production, i.e., when the organic food buyer is public. The remaining two papers focus on the retail market for organic food, i.e., when the organic food buyer is private. Specifically, the thesis adds to the empirical knowledge on GPP as an environmental policy instrument, using unique longitudinal data on organic food purchases in Swedish municipalities.

Further, using retail and wholesale level price data for organic and conventional food, the thesis adds to the knowledge on consumer demand and retailers' price setting behaviour, which can inform policy makers on market responses to changes in, e.g., organic subsidies.

In Paper [I], we empirically examine the relation between Swedish public sector's organic food purchases and the share of organic agricultural land, following a national GPP policy launched in Sweden in 2006. We find that organic food purchases can be associated with a significant positive effect on the share of organic farmland. This effect is mainly found on aggregated level and attributed to the joint purchasing power of Swedish municipalities. In Paper [II], we study the determinants of GPP uptake using longitudinal survey data from Swedish municipalities and controlling for selection bias. Results show that policies related to organic food purchases are, despite being voluntary, in fact implemented and contribute to an increase in organic food purchases. In Paper [III], I focus on private consumers and a specific food product, milk, and estimate own- and cross-price elasticities of demand for organic and conventional milk using Swedish retail scanner data. Results indicate a more price elastic demand for organic milk compared to conventional milk, and strong consumer preferences towards branded milk. In Paper [IV], I analyse wholesale to retail price transmission for organic and conventional milk products, using Swedish data and time-series analysis. Results show that the price transmission for conventional milk is characterized by both long-run and short-run asymmetries, whereas for organic milk, the long-run asymmetry is less pronounced.

The remaining part of the introduction presents the market for organic food in Sweden. This background is followed by a description of GPP as an environmental policy instrument, and consumer demand and price transmission within the market for (organic) food. Finally, I present a summary of each paper.

1.1 Background – Swedish market for organic food

Rules on organic food production and labelling have existed on EU level since the early 1990's. In brief, organic production constitutes a system of farm management and food production that combines best environmental practices, high animal welfare and a high degree of bio-diversity and self-reliance. This includes, e.g., crop-rotation, organically fed livestock, and restricted use of off-farm and chemically synthesized inputs (European Commission, 2007). For a food product to use the EU organic label, production needs to comply with regulation EU 2018/848 (repealing EC 834/2007). Regional labels often co-exist, and Swedish organic producers are usually also certified under the KRAV label (Swedish Parliament, 2010).

The share of organic agricultural land within the EU has steadily increased during the last two decades, although from small numbers. In 2017, the organic share of agricultural land within the EU

reached 7%, which represents a 70% growth over just the last decade (European Commission, 2019). Sweden ranks sixth within the EU in terms of organic hectares, and the share of agricultural land in Sweden amounted to about 19% in 2017. This puts Sweden, together with Austria (23%) and Estonia (20%), among the top three within the EU, and top five in the world (Willer & Lernoud, 2019).

Organic food products are more costly to produce compared to conventional products. These cost differences mainly stem from lower productivity in organic farming due to, e.g., reductions in agro-chemical inputs, lower livestock density, and higher labour intensity, but also from initial costs of conversion, more fluctuating yields and higher costs in later stages of the supply chain (Furemar, 2004; Jørgensen, 2012; Tranter et al., 2009). Within the EU, subsidies are paid out to compensate organic farmers for the public good they provide in terms of environmental benefits and increased animal welfare. Subsidy levels and disbursement rules differ between countries, types of production, and over time, which makes comparisons difficult.

In addition to organic subsidies, the organic farmer may also receive compensation from the market in the form of an organic price premium, which differs between products, regions and over time. For example, Furemar (2004) compares organic and conventional prices in five EU member states and finds an average organic price premium of about 40% in Sweden and Denmark, and above 60% in Germany and France. Organic price premiums, market shares and distribution channels are often considered to be inter-related. As organic markets mature, supply tends to catch up, and retail sales of organic food products are increasingly moved from high-priced specialized organic food stores to general retailers, resulting in lower price premiums and larger organic market shares (European Commission, 2005; Furemar, 2004; Hamm et al., 2002). General retailers represent about 90% of the organic retail sales in Sweden, with similar shares in Denmark and Great Britain, suggesting that organic products are widely accessible for consumers in these countries. Although organic retail sales are increasingly dominated by general retailers within the EU, specialized organic food stores are still prominent distribution channels in countries such as France, Germany and Italy (European Commission, 2005; Willer & Lernoud, 2019).

In 2017, the organic share of food retail sales in Sweden was 9.1%, which was the second largest in the world after Denmark (13.3%). Organic expenditures per capita in Sweden were the third highest, and amounted to EUR 237 (Willer & Lernoud, 2019). Organic retail shares tend to be largest for fresh fruit and vegetables, milk, and eggs, and this also holds for Sweden (Willer & Lernoud, 2019). Official data on organic imports and exports are not available, although estimates suggest that the imported share of organic food consumed has increased from 20% in 2003 to 57% in 2016, and that 1% of Swedish food exports is organic (Jørgensen, 2012; KRAV, 2016). This makes Sweden a net importer of organic food, like most EU countries (European Commission, 2010).

The Swedish government has formulated several targets on organic production. In 2006, the government set a target of 20% organic agricultural land by 2010, and 25% public organic food purchases by the same year (Government communication, 2006). None of these targets were reached by 2010. In the national food strategy decided in 2017, targets were updated and stated that 30% of the Swedish agricultural land, and 60% of public food purchases should be organic by 2030 (Swedish Government, 2017). In 2017, organic food purchases within the Swedish public sector (including municipalities and county councils) amounted to about 36%, and the share of organic land amounted to about 19% (Ekoweb.nu, 2018; Willer & Lernoud, 2019). By encouraging the public sector to increase their organic food purchases, Sweden's GPP policy can be seen as demand side-oriented, adding to direct agricultural policy in the form of subsidies, by having the market bear a larger share of the costs faced by farmers converting to organic production (Government communication, 2006).

1.2 GPP as an environmental policy instrument

Public sector purchases generally account for a substantial share of a country's GDP. For example, in 2015, the average share of GDP spent by public authorities on goods, services and works within the EU was 13% (European Commission, 2016). By use of this potential purchasing power, public procurement can provide the public sector an instrument of policy to attain political objectives, related to, e.g., labour standards and racial equality (McCrudden, 2004), or promotion of disadvantaged firms (Krasnokutskaya & Seim, 2011). The last two decades have seen a large increase in the use of GPP, which is defined as a purchasing process where the public authority strives to procure goods, services and works with less environmental impact, based on life cycle costs, compared to the non-green alternative (European Commission, 2008). The idea behind GPP is that the public sector uses its purchasing power to influence consumers and producers to alter the market in favour of more environmentally friendly products, thereby contributing to environmental goals on different levels (Edler & Georghiou, 2007; European Commission, 2008; Li & Geiser, 2005; Marron, 2003). GPP is thus part of the broader concept Sustainable Public Procurement, which takes social, economic, and environmental considerations into account in the procurement process (Brammer & Walker, 2011). Regional and national GPP initiatives are largely promoted and employed within the EU, and also within the OECD, with initiatives launched in, e.g., the US, Japan, and Australia (European Commission, 2008). However, despite clear recommendations from the European Commission to apply GPP to stimulate organic food production, there are few other countries that have launched a similar organic food policy as Sweden, Denmark being one exception (European Commission, 2019).

Although a relatively new area of research (Testa et al., 2012), the body of GPP literature has steadily increased during the last two decades (Cheng et al., 2018). In a recent literature overview,

Cheng et al. (2018) find that most studies on GPP relate to GPP practices and uptake, and that the effects of GPP is generally an understudied topic.

Although frequently used and advocated for, the effectiveness of GPP as an instrument of environmental policy is not self-evident. Environmental policy instruments can roughly be divided into command-and-control (e.g., uniform quotas, regulations, or certain performance standards), and economic, i.e., market-based, instruments (e.g., emission taxes, abatement subsidies, and tradeable permits) (Hanley et al., 2016; Tietenberg, 1990). GPP generally takes the form of green criteria stipulated within the selection and award process, and is therefore best described as a command-and-control instrument (Lundberg & Marklund, 2013).

When evaluating an environmental policy instrument, one often refers to its cost-effectiveness and objective-effectiveness (Hanley et al., 2016). The former relates to the extent to which the reduction in environmental damage is achieved at lowest possible societal cost, and the latter refers to the extent to which a policy instrument achieves its target outcome. With economic instruments, the cost of the environmental damage is meant to be accounted for in the producer's marginal cost, and each producer then chooses its profit-maximizing level of emission reduction, thus rendering a cost-effective outcome. The command-and-control instrument, on the other hand, is inflexible and will not result in a cost-effective outcome if the marginal cost of emission reduction differs among producers.

Theoretical work points to GPP policies being neither cost-effective, nor objective-effective. By its resemblance to a command-and-control instrument, Lundberg and Marklund (2013) argue that GPP cannot be considered cost-effective if the potential suppliers are heterogenous in their production technology, and therefore heterogenous with respect to marginal costs of adapting to the green criteria stipulated in the call for tender. This follows from principles of non-discrimination and equal treatment in the EU procurement directives, which prohibit procuring authorities to customize environmental criteria according to supplier-specific marginal costs of adjustment. Moreover, Lundberg et al. (2016) and Lundberg et al. (2015) argue that since participation in the tendering process (and adaptation to environmental criteria) is optional, GPP's objective-effectiveness is limited. In line with theoretical work by Marron (1997), Lundberg et al. (2016) also find that GPP policies can even have a counteracting effect on emissions. This follows from a situation in which the public sector substitutes conventional products with less polluting ones. With upward sloping supply curves, this leads to, under certain assumptions, a price increase of the green product, which, all else being equal, sparks a counteracting effect from private consumers and increases emissions.

However, despite limited cost- and objective-efficiency, GPP may, under certain market conditions, contribute to reduced environmental impact. Assuming a green product and a non-green (brown) substitute, Marron (1997, 2003), and Lundberg et al. (2016) predict that the potential of GPP

to reduce brown production and increase green production increases when (i) the public sector is a large buyer of both the brown and green products, (ii) the supply of both products is inelastic, and (iii) the private demand is inelastic for both products. One must therefore consider the characteristics of the specific market when analysing the potential impact of GPP. For example, Simcoe and Toffel (2014) study California's construction sector, where municipalities are a large buyer, and find that GPP can solve a coordination problem by facilitating the diffusion of green building standards within the private sector. Rietbergen and Blok (2013) study a GPP scheme used within the Dutch infrastructure and railway sector, where the public sector has substantial purchasing power, and find that the scheme can reduce yearly CO₂ emissions by 0.8–1.5% among its participants.

Turning to the Swedish organic food policy from 2006, the potential for GPP to contribute to an increase in the share of organic farmland may, at a glance, seem weak, since the public sector accounts for only 4% of the total market for food (Swedish Competition Authority, 2011). However, concentration is high among the wholesalers catering to the public sector, and public authorities make up a large part of the demand aimed at the two leading firms (Martin & Servera, 2017; Menigo Foodservice AB, 2017; Swedish Competition Authority, 2018b). This would potentially provide the public sector with necessary buyer power. Regarding (ii), Jørgensen (2012) points out that a Swedish GPP policy favouring organic food is likely to be hampered by short-term inelasticities in the supply chain, due to limitations in organic production related to machinery, knowhow, and the arable land itself. In terms of (iii), demand elasticities for organic and conventional food are not extensively studied in a Swedish context. However, studies from, e.g., the US (Glaser & Thompson, 1999), Denmark (Wier & Smed, 2000), and the Netherlands (Bunte et al., 2007) indicate, in general, a relatively more price elastic demand for organic food compared to the often unit elastic demand for conventional food.

An important aspect of the feasibility of GPP is that GPP policies are often voluntary to implement and carry out (European Commission, 2012; Michelsen & de Boer, 2009), and often involve an expected increase in costs (Bouwer et al., 2006). Public procurement is, at the same time, largely decentralized and carried out by public servants whose capacity to implement GPP is balanced against political goals, financial constraints, and market specific conditions (Gains et al., 2004). Among the studies addressing the determinants of the use of GPP, i.e., GPP uptake, results show that the main barriers of successfully implementing GPP are (i) insufficient financial resources, (ii) lack of GPP and environmental knowledge, (iii) lack of organisational goals and structure, and (iv) lack of political commitment (Brammer & Walker, 2011; Cheng et al., 2018; Michelsen & de Boer, 2009; Testa et al., 2012). The barriers identified, and the voluntary and decentralized aspects of GPP, suggest that the mere existence or adoption of a GPP policy does not necessarily mean that it will be carried out

or result in more environmentally friendly purchases. To my knowledge, there are no previous studies that treat political ambitions or GPP uptake as continuous variables, nor are there any studies that consider, specifically, policies related to organic food purchases. Purchasing an organic food product instead of a conventional one is indeed, all else being equal, more costly for a public authority. Jørgensen (2012) compares a conventional and organic basket of foods and finds an average organic price premium of 66%.

To summarize, the assessment of whether GPP is to be considered an appropriate and feasible environmental policy instrument can be made from three dimensions: (i) the criteria against which the policy is to be evaluated, (ii) market specific conditions for GPP to impact the environmental target at hand, and (iii) the capacity of the procuring authority to take environmental considerations in the call for tender. In Paper [I], we address (i) and (ii), and in Paper [II], we address (iii).

1.3 Organic food - Consumer demand and price transmission

In Sweden, private consumption (excluding restaurants) makes up about 87% of the total food market (Ekoweb.nu, 2018). While GPP focuses on public purchases, governments are aware that, in the case of organic food, private consumption must also increase if targets on increased organic production are to be reached (European Commission, 2004; Government communication, 2006). Subsidies aimed at organic production are motivated by organic producers providing society with a public good, but consumer demand and price-setting strategies of actors within the supply chain are pivotal for subsidies to have the intended effect. For example, if organic subsidies increase conversion to organic production without there being any demand for organic food, these products will either be sold as conventional food products, or at a heavily reduced price premium, both of which options reduce profitability and partially counteract the subsidy's effect (Jørgensen, 2001). While an increase in the level of organic subsidy is likely to reduce the organic price premium and increase demand, large increases in subsidy levels may also lead to market distortions. Knowing what the market demand looks like, and how consumers react to price changes, is thus central for decisions regarding the design of organic subsidies and other price altering policies, e.g., differentiated VAT rates for organic and conventional products or input factors.

Studies investigating price elasticities of demand typically employ retail scanner data (see, e.g., Bunte et al. (2007)) or household scanner data (see, e.g., Schröck (2012)). In general, studies show that consumers have a relatively more price elastic demand for organic food compared to conventional food, although magnitudes of price elasticities differ between countries, products and over time (Bunte et al., 2007; Glaser & Thompson, 1999; Wier & Smed, 2000). The price elasticity of demand is closely related to market characteristics, such as market shares, organic price premiums,

distribution channels, and search costs. A general conclusion is that demand for organic food is more price elastic when organic price premiums are large, compared to when premiums are small (Bunte et al., 2007; Glaser & Thompson, 1999; Schröck, 2012). Organic price premiums tend to be larger in markets where organic market shares are small, and organic food products are separately distributed and sold (Hamm et al., 2002).

Compared to many other countries, organic food products in Sweden are well represented in general retailers' assortments, and sales in specialized organic food stores are marginal. However, despite a large focus on organic consumption in Sweden, studies on the price elasticity of demand for organic food are lacking. Paper [III] in this thesis contributes to filling this gap, and also addresses the existence of private label products within the Swedish food market.

The price-setting strategies among actors within the supply-chain may also influence the potential of, e.g., organic subsidies or differentiated VAT rates. A change in the producer price generates different impacts on the price faced by the consumer, depending on the speed, magnitude and symmetry/asymmetry of price transmission. If asymmetry is present, positive and negative price changes are transmitted differently in terms of speed and magnitude. For example, positive asymmetry means that a price increase is transmitted faster or to a larger extent, compared to a price decrease. In the case of downstream price transmission, e.g., wholesale to retail, an upstream price increase will then lead to a larger, or faster change in the price faced by the consumers, than in the case of a price decrease. The magnitude of price transmission can be related to factors such as market power, price elasticities, and the extent to which a product is processed (McCorrison et al., 2001), whereas asymmetric price transmission tends to be related to market power or adjustment costs (Meyer & von Cramon-Taubadel, 2004). These factors may differ across markets and products. In Sweden, concentration levels within the food retail segment are substantial. In 2013, one retail chain accounted for a market share of about 50%, and the three largest chains had a joint market share of 86%, compared to, e.g., 60% in Germany, and 20% in Italy (Swedish Competition Authority, 2018a).

A majority of the previous studies on agricultural markets show evidence of positive asymmetric price transmission (Meyer & Von Cramon-Taubadel, 2004). However, studies are generally lacking for organic food. Due to organic and conventional food products being qualitatively differentiated, their price transmission may differ (Würriehausen et al., 2015), meaning that price shocks (due to, e.g., changes in subsidies, VAT rates) may affect organic and conventional markets differently. The aim of Paper [IV] is to address this gap in the literature. Moreover, by allowing for an asymmetric cointegrating relationship between prices, the analysis covers both short- and long-run asymmetries, the latter of which has been largely overlooked in previous studies of price transmission in agricultural markets.

2. Summary of the papers

Paper [I]: How Green Public Procurement can drive conversion of farmland: An empirical analysis of an organic food policy

In this paper, we study the relation between Swedish public sector's organic food purchases and the share of organic agricultural land. An overarching objective of the paper is to contribute to the assessment of GPP as an environmental policy tool. The research question departs from a policy on GPP decided by the Swedish Government in 2006, which states that organic food purchases among the Swedish public sector should increase in order to incentivise Swedish farmers to convert to organic production, thereby contributing to national environmental quality goals.

We use a rich set of panel data from the period 2003–2016 which includes information on municipalities' organic food purchases, land use in Swedish counties, and national subsidies aimed at organic production. Data on municipalities' yearly organic food purchases is sourced from a survey sent out by the independent organization Ekomatcentrum. This type of detailed purchasing information is rarely available, and this paper is the first, to our knowledge, to empirically analyse the relation between public organic food purchases and organic farmland.

Results show that public organic food purchases can be associated with a significant positive effect on organic farmland. This effect is mainly found on aggregated level and attributed to the joint purchasing power of Swedish municipalities. In other words, even though the Swedish public sector only accounts for 4% of the Swedish market for food, it appears as if uncoordinated public authorities together make up a sufficiently large buyer to reach purchasing power through oligopolistic wholesalers. In terms of a local effect, i.e., within the own county, we conclude that increased public organic food purchases within an individual county generally creates none, or small, incentives for producers within that county to convert to organic production. This is hardly surprising given the fact that production of (organic) food does not necessarily take place in the same county as where the purchasing public authority is located. Also, public authorities are largely prohibited to favour local suppliers. National agricultural subsidies aimed towards organic production also show a significant positive effect, in line with expectations. Overall, the findings of the reduced-form approach in this paper appear to be in line with what the theoretical literature on GPP states about requiring the public sector to be a relatively large buyer for environmental criteria to influence greening of the supply side.

Paper [II]: Green Public Procurement: An empirical analysis of the uptake of organic food policy

In this paper, we explore the determinants of GPP uptake, using panel data for 290 Swedish municipalities for the years 2003–2016. The study departs from a national GPP policy launched in 2006, which encouraged Swedish municipalities to increase their organic food purchases in accordance to locally decided, and financed, policy ambitions. As these types of policies are generally (i) costly to the public authority, (ii) voluntary to adopt and adhere to, and (iii) implemented at decentralized level among heterogeneous subjects, merely adopting a policy related to organic food purchases does not necessarily imply that the policy in question is successfully implemented.

This paper adds to the ample literature on determinants of GPP uptake and implementation in several ways. While previous studies reduce political goals and GPP uptake to binary variables, we use three different measures of a municipality's organic food policy, and treat GPP uptake as a continuous variable, measured in both percentage shares of organic food and organic expenditures per capita. The paper also adds significant value of scope with its relatively large coverage over municipalities and years, and for a specific product. To our knowledge, this is also the first empirical analysis to account for selection bias when using survey data on GPP.

By coupling a unique and extensive data set on municipalities' food purchases and organic food policies, with municipality characteristics, we perform a multivariate analysis of the relation between factors related to (i) policy, (ii) financial constraints, and (iii) GPP awareness, and GPP uptake. Estimations are performed using mainly OLS, while accounting for potential selection bias using Heckman's selection model, modified to fit panel data. The main finding is that political goals have a significant and positive effect on the share of organic food purchases, suggesting that there is an uptake and that the voluntary policy is in fact implemented. Secondly, the increase in expenditures per capita devoted to organic food is quite substantial, following the adoption of an organic food policy. An extensive robustness check is performed, by, e.g., including a time trend instead of year fixed effects, estimating a pooled Heckman model without accounting for unobserved heterogeneity, and using different instruments in the selection equation.

Paper [III]: The Swedish consumer market for organic and conventional milk: A demand system analysis

In this paper, I study demand for organic and conventional milk in the Swedish market, using weekly scanner data for the years 2011–2017. Increasing the organic production is becoming an important environmental target for many governments, and consumer demand for organic food is pivotal in reaching these targets. Despite a large policy focus in Sweden on organic production, there is scarce knowledge on the price sensitivity of consumers regarding organic and conventional food products.

Studying milk, specifically, is of interest as the milk sector constitutes a large agricultural sector in Sweden with relatively high organic conversion, and organic milk has a relatively large market share within the milk segment. The paper adds to previous studies of demand for organic and conventional milk, by (i) studying a market with relatively small organic price premiums, (ii) using a highly representative sample of retailers, and (iii) differentiating between private labels and brands.

Data on sales quantities and prices is retrieved from the market research company AC Nielsen, which collects weekly barcode level data from a large sample of stores, and aggregates it to six Swedish regions, encompassing all Swedish municipalities. The sample is divided into four categories, based on label (organic/conventional) and manufacturer (private label/brand). Private label milk products were introduced in the Swedish market quite recently, and their market shares in the sample are quite small. The organic price premium for the sample is, on average, 15.8% within the branded segment and 27% within the private label segment. Estimations are performed using a quadratic almost ideal demand system (Q-AIDS), which allows for goods to be luxury goods at one income level and necessities at other. I use income to instrument for endogenous expenditures and exploit the panel structure of the data when instrumenting for endogenous prices, by using prices in another region as instrumental variable.

Results show that demand for organic milk is more price elastic than for conventional milk, within the branded segment (-1.944 vs -1.053), but not within the private label segment (-2.278 vs -2.445). The magnitude of the own-price elasticity for organic branded milk is relatively large compared to previous studies, considering the low organic price premium in the Swedish sample. One possible explanation to this is that the local setting of milk brands in Sweden already appeals to consumers' sense of making sustainable purchasing decisions. Demand for private label products is more price elastic compared to branded products, both within the organic and conventional segment. This may be due to private labels' relatively late entrance in the Swedish market, following a long period of regional dairy monopolies, resulting in strong consumer preferences for regional milk brands.

Paper [IV]: Price transmission for organic and conventional milk products in Sweden

In this paper, I study wholesale to retail price transmission for organic and conventional milk in the Swedish milk sector using monthly price data from Jan 2007–Nov 2017, and time-series analysis. A large body of literature addresses the efficiency of food supply chains by studying price transmission, but there is little knowledge on price transmission for quality-differentiated food products, and particularly for organic food vis-à-vis conventional food. Using data from a market with similar distribution channels for organic and conventional food, this paper adds to this scarce literature.

The study of price transmission in the milk sector makes for an important case, as the milk sector constitutes a large agricultural sector in Sweden, with a relatively large share of organic production and consumption. Moreover, the Swedish dairy sector and food retail sector are characterized by high concentration levels, providing especially the retail level with considerable bargaining power. The effects of changes in subsidies or taxes on demand depend on how price changes are passed through the supply chain. From a policy and welfare perspective, it is thus important to consider price transmission for organic and conventional milk, when designing policies aimed at these markets.

The dataset employed covers monthly wholesale prices for organic and conventional 1.5% milk, provided by a large Swedish dairy, coupled with corresponding retail prices. The data on retail prices is provided by the market research company AC Nielsen, which collects barcode level data on, e.g., sales, prices, manufacturer, and organic label from a representative sample of stores for six regions. Since the wholesaler's products are not sampled in all regions, retail prices in this study are from only one region, which encompasses 45 Swedish municipalities.

Estimations are performed using the non-linear autoregressive distributed lag model (NARDL). Its main advantage is that it allows for asymmetric cointegration of prices. This enables a simultaneous analysis of asymmetries in the short-run dynamic error corrections (speed), and in the long-run cointegration relationship (magnitude), the latter of which has been largely overlooked in previous studies. Results indicate a positive asymmetric price transmission for conventional milk both in the long-run and the short-run. In the long-run, a 1% increase (decrease) in conventional wholesale price is associated with a 0.702–0.789% (0.426–0.595%) increase (decrease) in conventional retail price. For organic milk, results point to a less asymmetric long-run price transmission. Specifically, a 1% increase (decrease) in organic wholesale price is associated with a 0.856–0.908% (0.711–0.859%) increase (decrease) in organic retail price. Organic consumers will therefore experience smaller differences between surplus losses and gains following positive and negative wholesale price changes, respectively. Although the analysis does not include formal tests of what causes the asymmetry, some possible explanations are discussed, such as retailers' misuse of market power, differing margins and demand elasticities for organic and conventional milk, and lower organic search costs compared to markets with separate distribution channels for organic products.

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